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Unit 1 - Topi Investigation

PDF Documents

PDF Document 1: 1738215882977-Chapter 9 - Unit 1 - Topi Investigation.pdf

This PDF document is included in the unit materials.

UNIT 1 TOPI INVESTIGATION

Introduction

In this chapter, you will embark on an exciting journey to design and implement your own smart device that helps with tasks around your house. This self project will allow you to apply everything you have learned about programming, engineering, and innovation in a practical, hands-on way.

Why Self Projects are Important

Self projects are a great way to help you become an innovator. Here's why they are important:

BE CREATIVE

Self projects let you think of new and interesting ways to solve problems. This kind of creative thinking is important for coming up with new ideas and inventions.

SOLVE PROBLEMS

When you work on a self project, you learn how to find solutions to challenges. You'll practice figuring out what the problem is, thinking of different ways to fix it, and choosing the best solution. This helps you become a better problem solver.

LEARN BY DOING

Building a project helps you understand what you have learned in class. When you use your knowledge to make something real, it becomes easier to remember and more fun to learn.

BE INDEPENDENT AND CONFIDENT

Completing a self project means you have to take charge and work on your own. This helps you feel more confident in your abilities and teaches you to rely on yourself.

THINK LIKE AN ENGINEER

Self projects teach you how to design, build, test, and improve your ideas. These are the same steps engineers use to create new technology. This experience helps you understand how things are made and prepares you for future projects.

FOLLOW YOUR INTERESTS

Working on a project you care about keeps you excited and interested. You get to explore topics you like, which makes learning more enjoyable.

MAKE A DIFFERENCE

By creating a smart device to help with tasks in your house, you can see how your work makes life easier. It's rewarding to see your ideas help solve real problems.

Creations by Students Through Self Projects

Remote control Boat

- **MODI Modules:**
  Motor A, Motor B, Battery, and Network
- **Additional Materials:**
  Plastic food container, LEGO bricks
- **How It Works:**
  Using the joystick controller in the app "Let's MODI", the user controls the speed and direction of Motor A and Motor B. watering spray automatically waters the plant.

Smart Plant Watering System

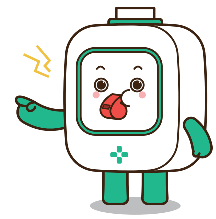
- **MODI Modules:**
  LED, Speaker, Display, Motor B, Button, Environment, and Network
- **Additional Materials:**
  Artificial leaves, paper, cardboard, etc.
- **How It Works:** The default status is that the display shows the current temperature and humidity levels. If the humidity level falls below a certain threshold, the LED and speaker provide visual and auditory alerts. Then, the watering spray automatically waters the plant.

Unit Overview

In Unit 1, you will explore and identify a suitable topic for your self project. This unit will guide you through the following processes:

1. IDENTIFYING NEEDS IN YOUR HOUSE

- Think about different tasks or activities in your house that could be improved with a smart device.

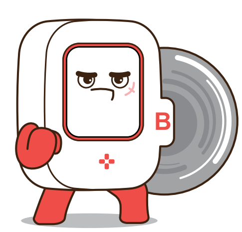


2. RESEARCHING EXISTING SMART DEVICE

- Look into existing smart devices for inspiration.
- Note what problems they solve, how they work, and any useful features.

3. CHOOSING YOUR PROJECT TOPIC

- Select a specific topic for your project based on your research.
- Consider the feasibility, impact, and your interest in the idea.



4. QUESTIONING THE DETAILS

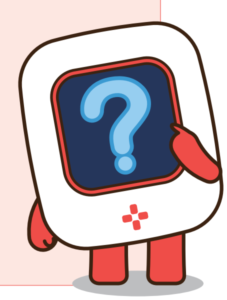
- Answer key questions about the tasks to be improved, necessary features, existing devices for inspiration, and required resources.

Identifying Needs in Your House

Think about the different tasks or activities in your house that could be made easier or more efficient with the help of a smart device. It could be anything from automatic lighting to a device that helps with cleaning.

YOUR THOUGHTS AND IDEAS

- Write down specific tasks or activities in your house that you think could be improved with a smart device. Be detailed about the problems you want to solve and how a smart device could help.



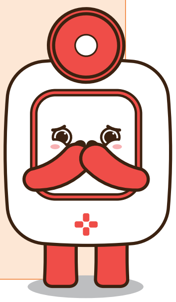
Researching Existing Smart Devices

Look into existing smart devices to get inspiration. What problems do they solve? How do they work? This research will help you understand what is possible and spark ideas for your own project.

Consider the technology behind these devices: how do they sense, process, and respond to different inputs? For example, a smart thermostat adjusts the temperature based on user preferences and external conditions, while a robotic vacuum cleaner navigates and cleans the floors autonomously.

YOUR THOUGHTS AND IDEAS

- Write down the smart devices you find interesting. Note what problems they solve, how they work, and any features you think are particularly useful.



Choosing Your Project Topic

Based on your investigation and research, choose a specific topic for your project. Consider the feasibility, impact, and how much you are interested in the idea. Your topic should focus on creating a smart device that helps in your house.

Consider how you can realize your project using MODI modules. For example, if your project involves automatic lighting, think about using sensors to detect movement and MODI modules to control the lights. If it's a smart cleaning device, consider how motors and sensors can work together to automate the cleaning process.

YOUR THOUGHTS AND IDEAS

- Write down the smart devices you find interesting. Note what problems they solve, how they work, and any features you think are particularly useful.

Questioning the Details

What tasks in your house could be improved with a smart device?

- Identify daily activities that are time-consuming, repetitive, or could benefit from automation, such as cleaning, organizing, security, or managing appliances.

Are there any existing smart devices that could inspire your project?

- Look at similar smart devices on the market. Analyze their features, strengths, and weaknesses to incorporate or improve upon these ideas in your project.

What resources will you need to complete your project?

- List the materials, tools, and knowledge required, including MODI modules, sensors, programming skills, and support from teachers or family members.

Think About It

Reflect on the various activities in your home and how they could be improved with smart technology. Discuss your ideas with your classmates and write down potential topics for your project.

YOUR THOUGHTS AND IDEAS

- Write down your answers to these questions. Drawing with writing is fine to help illustrate your ideas and make your notes more detailed and clear.

Conclusion

Summary of Findings

Summarize the key points you have discovered during your investigation. Reflect on the tasks in your house that could be improved with a smart device, the existing devices that inspired you, and the resources you will need.

YOUR ANSWER

Key Features

List the key features that will make your smart device effective. Describe how these features will address the tasks or problems you identified in your house.

YOUR ANSWER

Required MODI Modules

Identify the MODI modules you plan to use in your project. Explain the functions of each module and how they will work together to create your smart device.

YOUR ANSWER